

Watch-Out: Real-Time Tracking and Assistance for Wandering Alzheimer’s Patients Using Wearable Technology

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Abstract

Wandering poses a critical safety risk for individuals with Alzheimer’s disease, leading to injuries, fatalities, and immense caregiver stress. This paper introduces Watch-Out, a comprehensive safety system combining a smartwatch app, an NFC-enabled smart door lock, and a caregiver mobile app to address wandering behaviors. The system employs NFC tagging and voice prompts to ensure cognitive engagement during exits, reducing the likelihood of unsupervised wandering. Features include real-time GPS tracking, geofencing alerts, health monitoring, and community-based emergency response mechanisms. Grounded in user-centered design and insights from caregivers and experts, Watch-Out enhances patient autonomy while alleviating caregiver burdens. By integrating technology with empathy, the system ensures safety and dignity for patients while fostering peace of mind for caregivers. This solution redefines Alzheimer’s care, paving the way for safer, more supportive communities.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing; Ubiquitous and mobile computing systems and tools.

Keywords

Alzheimer’s Disease, Wandering Prevention, Caregiver Support, GPS Tracking, Geofencing, Community Engagement

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1 INTRODUCTION

Wandering is a serious safety risk for individuals with Alzheimer’s disease, often leading to injuries, exploitation, or fatalities. Recently, four individuals with dementia in Georgia were found dead after wandering from home, emphasizing the urgency of addressing this issue [1]. Advanced dementia increases the risk of harm to patients and others [2], while families face constant anxiety, and emergency services are heavily burdened. Addressing wandering aligns with global goals like SDG 3 (Good Health and Well-being) and SDG 11 (Sustainable Cities and Communities) [3], emphasizing the need for safer, more sustainable communities [4]. Current solutions are often ineffective or intrusive, with GPS trackers failing if not carried [5]. Caregivers face immense emotional strain due to the lack of reliable tools [6]. Watch-Out addresses this by integrating a smartwatch with a smart door lock requiring NFC tagging for exits, promoting safety and cognitive engagement. Key features include real-time notifications, navigation tools, geofencing, and community-based support, ensuring timely intervention. This system reduces caregiver stress while empowering patients to maintain dignity and autonomy. Grounded in user research, expert consultations, and caregiver insights, Watch-Out employs user-centered design to address real-world challenges. This paper outlines its development, demonstrating how it enhances safety and quality of life for individuals with Alzheimer’s and their caregivers.

2 BACKGROUND AND DESIGN APPROACH

2.1 Wandering in Alzheimer’s

Wandering is a significant issue for individuals with Alzheimer’s disease, affecting safety and well-being. Alzheimer’s is a neurodegenerative disorder impairing memory, cognition, and behavior [7], accounting for 60-70% of dementia cases globally [8]. As the disease progresses, patients struggle with orientation, familiar places, and judgment, leading to wandering. Six in 10 individuals with dementia wander [9], and 74% leave home using walking, biking, driving, or public transport. Tragically, 30% of cases result in death due to exposure, drowning, or accidents, with survival rates dropping to 20% after three to five days [4]. Wandering risk increases with disease progression. In the mild stage, confusion is brief [10], while the moderate stage involves significant disorientation and a

desire to “go home” [11]. Severe stages bring loss of communication and awareness, heightening wandering risks. Caregivers face emotional and physical burdens balancing patient safety and quality of life. Solutions like environmental changes, wearable devices, and routines help but remain insufficient. A multifaceted approach is essential to ensure safety and dignity for Alzheimer’s patients.

2.2 Expert Consultations

Through discussions with the CEO of an elderly care institute, significant challenges related to Alzheimer’s patients and their caregivers were identified. Wandering behavior creates risks for patients, caregivers, and the community, requiring constant supervision and posing legal liabilities for care facilities. The CEO highlighted the emotional burden on caregivers, who experience constant anxiety about patient safety and the strain of continuous monitoring. Families also face ongoing worry while managing caregiving demands. Many facilities use tracking and monitoring solutions to mitigate risks, but these are often ineffective if patients refuse or remove them. As a result, some facilities resort to locking doors, ensuring safety but negatively affecting patients’ mental and emotional well-being, leading to feelings of isolation, agitation, and depression. The CEO shared a case where a patient left home undetected, entered a neighbor’s residence, and caused disturbances. Such incidents highlight the safety risks posed to both patients and the community, emphasizing the critical need for preventive measures.

2.3 Interviews with Caretakers

In an interview with a caregiver for his mother with Alzheimer’s, he described constant anxiety about her potential wandering and safety. His main concerns were centered on her entering unfamiliar areas or getting lost. He shared, “The fear of her wandering off and not knowing where she is is constant.” The dual responsibility of maintaining his job while providing care has significantly impacted on his personal life, causing him to miss important family moments. He expressed a feeling perpetually torn between these obligations, stating, “Balancing my job and taking care of her is exhausting.”

Despite the challenges, the caregiver expressed optimism about potential solutions. He believed that real-time location tracking with alerts and a smart lock system could be “a lifesaver,” potentially reducing his stress while ensuring his mother’s safety. The prospect of an integrated system combining wearable devices with smart home technology particularly appealed to him as a comprehensive solution.

2.4 Current Challenges with Existing Solutions

Current technological solutions for Alzheimer’s patients offer basic safety measures but fail to address the complex neurobiological and behavioral challenges of the disease. A key limitation is their inability to manage hippocampal deterioration, which significantly affects spatial memory and navigation. Research shows that over 45% of wandering incidents occur in previously familiar locations, reflecting a progressive decline in the brain’s ability to maintain spatial representations rather than simple disorientation [12]. Additionally, passive tracking systems like GPS watches can negatively

impact psychological well-being. Studies reveal these systems foster “learned helplessness,” where patients, aware of being monitored, gradually lose their navigational abilities [13]. Emotional well-being, closely tied to spatial cognition, is another critical factor often neglected. Research shows that engaging in emotionally stimulating activities (e.g., adventure, fun) raises cortisol levels in 82% of cases, potentially accelerating cognitive decline by affecting hippocampal function [14, 15].

2.5 Key Insights for Solution Development

Through a comprehensive review of expert consultations, caregiver interviews, and an analysis of existing solutions, several critical insights have emerged to develop a solution to address wandering in Alzheimer’s patients. The uniqueness of our idea—providing more personal autonomy and social integration for patients rather than simple isolation—has been organized as seen in Table 1.

3 DESIGN OF WATCH-OUT

3.1 Overview

Building on insights from expert consultations, caregiver interviews, and an analysis of current challenges, the “Watch-Out” system offers a comprehensive solution to address wandering in Alzheimer’s patients. This integrated system combines a smart NFC-enabled door lock, a mobile caregiver app, and a smartwatch app to provide real-time tracking, notifications, and support. The door lock requires patients to touch against the integrated NFC tag with their smartwatch before exiting, promoting awareness of their movements while reducing unsupervised wandering. The smartwatch delivers real-time updates to caregivers, alleviating the stress of constant supervision and enabling proactive responses through features like GPS tracking, geofencing alerts, help requests, and a digital compass. This seamless integration ensures safety, autonomy, and peace of mind for both patients and caregivers.

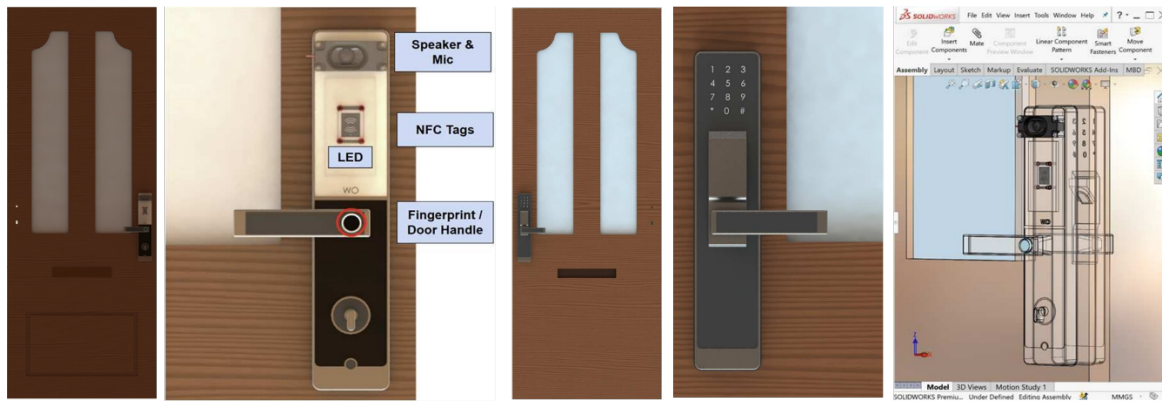
3.2 Door Lock Mechanisms and Functionality

The Watch-Out system integrates a smart NFC-enabled door lock designed to serve as a gatekeeper, offering structured and safe exit points for patients. The door lock system, as shown in Figure 1, is equipped with an RFID/NFC reader and LED guidance for usability and feedback. Its design ensures that patients interact with the system meaningfully before exiting, which reduces wandering risks. By providing a structured yet accessible exit process, the Watch-Out door lock upholds patient autonomy while maintaining safety. Unlike traditional locks that impose strict restrictions or no exit except caretakers. This system offers a monitored but viable means of exiting, through answering simple questions and tagging the smartwatch to the door.

According to Figure 1a, the door includes interactive elements, such as a speaker, to engage patients with structured dialogue. This feature reinforces cognitive engagement by asking about their destination or clothing choices, grounding them in the present moment, and reducing wandering-related confusion [16]. Verbal interactions are designed for clarity, using short and simple sentences, calm tones, and average speaking speeds for ease of understanding [17, 18]. Figure 2 illustrates the system’s operation, highlighting a dual-verification process that combines physical interaction with

Table 1: Insights and Solutions

Insights	Description	Possible Solutions
Emotional Well-being	Alzheimer's patients exhibit better spatial orientation and overall safety when they feel emotionally secure and engaged. This challenges the traditional focus on location monitoring.	Incorporating features that foster emotional engagement. Such features could include gentle guidance (navigation with speakers), help from others, etc.
Caregiver Burden	Caregivers experience significant emotional and physical burdens associated with managing wandering behaviors.	Real-time monitoring, alerts, and intervention tools.
Community Engagement	Caregivers are often unable to be present when a patient goes missing, especially in public spaces or when the patient has ventured far from home. “A better chance of survival/being found in urban environments because of a higher probability of a “good Samaritan” stepping in” [4]	Leveraging the support of the community becomes crucial to ensuring the safety and timely recovery of wandering individuals.

**Figure 1: (a) Inside Door 3D Model with smart door lock installed. (b) Outside Door 3D Model (c) Modeling Process**

the smartwatch and cognitive engagement through voice prompts. This ensures patients are equipped with their safety devices and provides caregivers with real-time notifications, including the patient's destination, exit time, and responses. Caregivers can also bypass these measures using a fingerprint lock when necessary.

Overall, the Watch-Out door lock acts as a gatekeeper, utilizing an RFID/NFC reader and LED guidance to direct the patient where to tag the smartwatch. The speaker initiates meaningful dialogue and asks purposeful questions. These “conscious exits” require both physical interaction via the smartwatch and voice-based cognitive engagement, ensuring patients are prepared for independent movement. Caregivers receive timely alerts for proactive monitoring, with a fingerprint lock available for quick bypasses when needed.

3.3 Watch-Out Mobile App for Caregivers

Once the Alzheimer's patient has successfully worn their smartwatch and exited the house, caregivers can view the patients

through Safe Care Tracker application—integrating location monitoring, health tracking, and emergency response into a cohesive safety framework for patients with Alzheimer's disease.

The application starts with a real-time location system—providing continuous tracking through an interactive map interface, enhanced by customizable geofencing utilities (see Figure 3b). Caregivers can create precise safe zones by drawing boundaries directly on the map, whether defining broad areas like a “home wide” radius of 175km or specific locations such as a frequently visited supermarket. When a patient leaves these virtual boundaries, the system immediately alerts caregivers, providing crucial early warning of potential wandering incidents. This geofencing feature works in tandem with historical movement analysis, generating heatmaps that reveal patterns in patient behavior and identifying high-risk areas or common wandering paths.

Secondly, health monitoring through smartwatch integration provides continuous vital sign tracking, displaying real-time data including blood pressure (shown as 127 mmHg) and heart rate (88

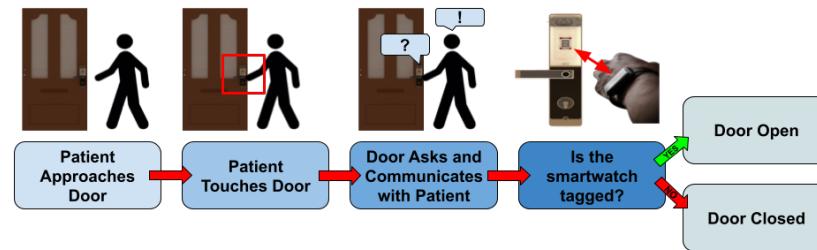


Figure 2: Flowchart illustrating the operational process of the door lock, including patient interaction, cognitive engagement, and verification steps to ensure safe and monitored exits.

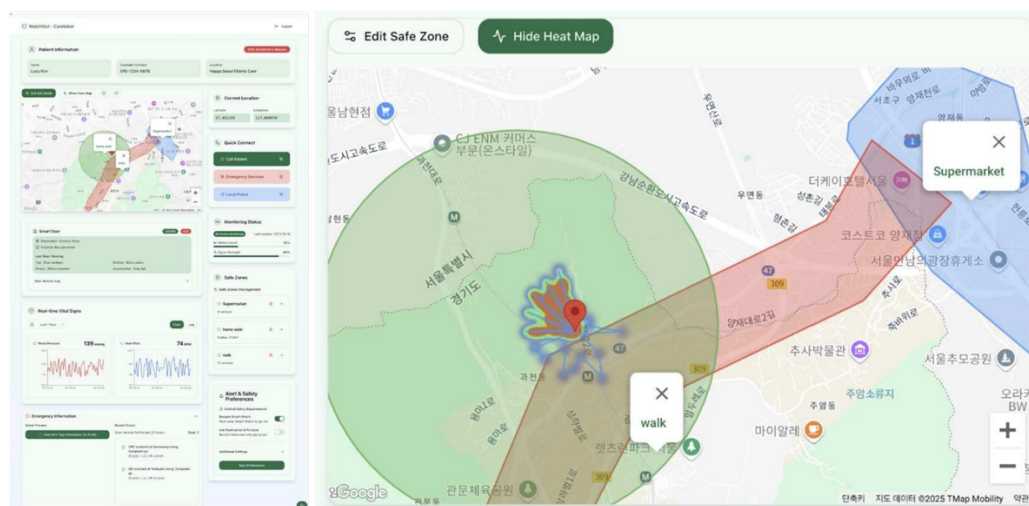


Figure 3: (a) Full view of website/application. (b) View of set safe zones.

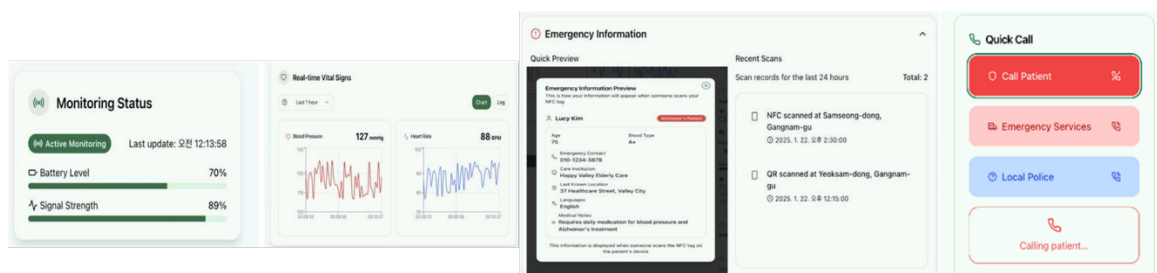


Figure 4: (a) Battery and signal levels. (b) Vital signs. (c) Emergency scan history and information (d) Quick Call

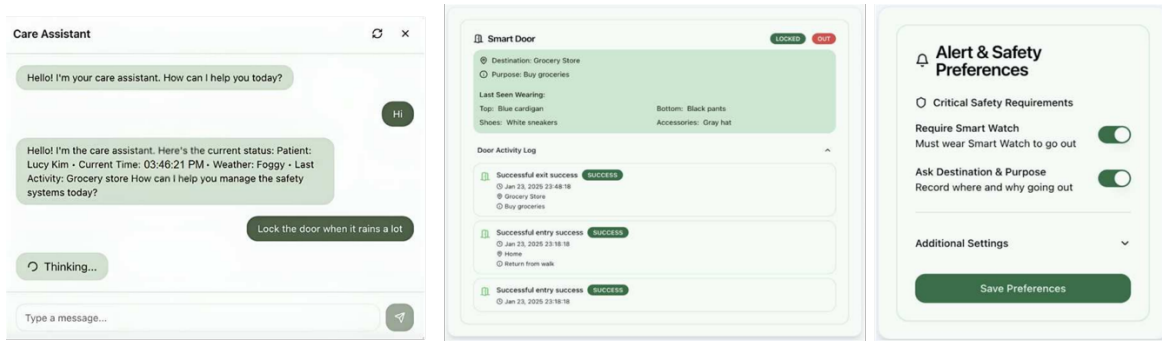


Figure 5: (a) Care Assistant. (b) Smart door lock data

bpm) through clear graphical interfaces (see Figure 4b). The system maintains transparent device status monitoring, with battery levels (70%) and signal strength (89%) consistently displayed to ensure reliable operation (see Figure 4a). This data triggers automated alerts to caretakers when concerning patterns emerge. This predictive capability allows for early intervention before any critical situation develops.

Thirdly, the emergency response framework uses a mechanism called “Quick Scan.” By touching an external electronic device to the patient’s smartwatch, the system provides immediate access to emergency services while simultaneously transmitting critical information (see Figure 4c). The NFC/QR “tag” code integration allows first responders to instantly access essential patient data, even if the patient is unable to communicate. This interface allows emergency communication through clearly defined channels for contacting the patient, emergency services, or local authorities. In addition, caregivers can control what kind of information they want to receive. Alert customization enables precise control over notification parameters, ensuring critical information reaches caregivers through their preferred communication methods. The system allows for time-based restrictions and risk-level adjustments, minimizing alert fatigue while ensuring appropriate responses for serious situations.

Next, the Quick Call feature enables caregivers to establish immediate, hands-free communication with the patient through the smartwatch’s built-in speaker, bypassing the need for the patient to accept or decline the call (see Figure 4d). This functionality is particularly critical in emergency situations, where rapid intervention can prevent harm, and in scenarios where the patient may be unable to interact with the device but can still receive auditory guidance. By ensuring an uninterrupted communication channel, Quick Call facilitates real-time caregiver intervention, enhances patient safety, and provides reassurance in critical moments, making it a vital component of the system’s emergency response framework.

Furthermore, an integrated chatbot assistant (“Care Assistant” in Figure 5a) provides guidance for system configuration and commands, helping caregivers optimize settings for their specific needs as well. Lastly, the smart door lock transfers data to this app (Figure 5b). Lastly, the smart door lock is connected to the app by transferring data (Figure 5b). When the patient answers a series of questions, the information is transmitted to the smartwatch application. Additionally, the system incorporates weather-based



Figure 6: (a) Digital compass with voice guidance. (b) Airdrop emergency information and ask for help.

safety protocols, automatically restricting exits during hazardous conditions like extreme temperatures or severe weather.

3.4 Watch-Out Smartwatch App for Patients

As the patient leaves the house with the smartwatch worn, the smartwatch app activates. The watch app interface is designed with a minimalistic approach to prioritize ease of use for elderly individuals [19] (see Figure 6). While comprehensive data is relayed to caregivers, the user-facing interface focuses on essential features. A primary function is the digital compass, which vibrates and provides turn-by-turn guidance to direct users back to safety when they exit the safe zone. In addition, voice guidance has been implemented to enhance understanding and reduce cognitive load. In case the user ignores navigation prompts, the watch notifies caregivers and, if necessary, local authorities. Our digital compass function is designed to help patients remain within a safe and accessible area, ensuring caretakers can reach them easily while reducing the risk of accidents in distant or hard-to-reach locations. This feature stands out by adding safety and ethical considerations within a familiar GPS system by allowing patients a degree of freedom while ensuring their well-being.

Beyond direct caregiver-patient interaction, the watch also engages nearby individuals when necessary. If the user leaves the safe zone and disregards navigation, the watch can transmit essential information (e.g., caregiver contact details, clothes the person is

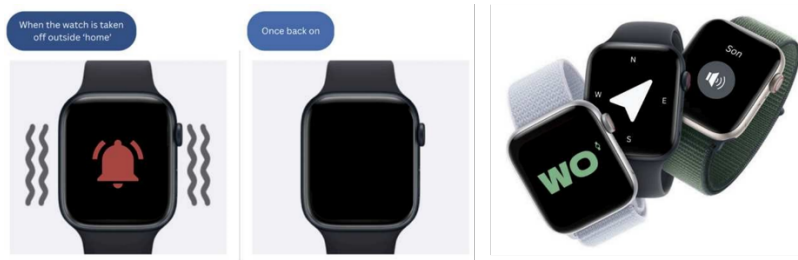


Figure 7: (a) Alarm in case of removal. (b) Watch-Out Logo, digital compass, and quick connect.

Table 2: Features, Previous Work, Our Solutions, and Uniqueness

Features	Previous Works	Our Solutions	Uniqueness
Patient Home-Leaving Restrictions	Most door-related products restrict autonomy, leading to isolation. They include i) double-locked doors with keypads and NFC tags for caretakers, ii) doors disguised as walls, and iii) manual metal locks.	Our system allows a viable way of leaving through two steps: 1) answering questions (destination, clothing) the smart door lock asks, 2) requiring tagging the smartwatch on to the NFC area.	Rather than enforcing isolation or neglect, our system provides a balanced approach, allowing patients to exercise rational decision-making and autonomy in leaving the house.
Assurance of Safe Return / Patient Danger Detection	Some smartwatch products have a simple GPS tracker, and a feature to call law enforcements to find the patient in need.	As the patient leaves the designated safe zones, the smart watch turns on digital compass mode, where a large arrow, vibrations, and audio instructions guide the patient to return to the safe zones.	Unlike basic GPS tracking, our system incorporates a digital compass, minimizing reliance on law enforcement and providing navigational cues for independent return to safety.
Assurance of Patients Wearing Wearables (Without Losing them, Breaking, etc.)	Most smartwatch products make caretakers to strongly fix the wearables on the patients—for permanent or long-time use.	Requiring a smartwatch at the door encourages patients to keep it, while removal outside triggers an immediate alert to caretakers.	Instead of constant wearable use, patients wear them only when going outside via NFC verification, reducing discomfort. Alerts prevent device abandonment.
External Help from People Outside Caretakers	Calling the police, law enforcement, or healthcare services. Also, when a patient is lost and can't be found, a public alarm is sent to the entire city to help find the person.	The smartwatch sends an "SOS" message with the patient's details (appearance, caregiver contact, etc.) to individuals within a 10m radius via AirDrop/Quick Share for help. Medical personnel can use QuickScan for personal and medical information.	Unlike broad city-wide alerts, our smartwatch sends emergency messages to nearby individuals who can help for more awareness and a higher chance of help. QuickScan can be used by medical personnel during emergencies for quick and efficient identification .

wearing, actions or words that trigger the patient) to nearby devices via AirDrop (Apple Watch) or QuickShare (Galaxy Watch). This enables bystanders to assist in guiding the user back or seeking appropriate support while ensuring public awareness in cases involving aggressive behavior.

To prevent the removal of the device, an alarm system is integrated as in Figure 7. If the watch detects a zero-heart rate outside the designated home zone, it triggers an alert prompting the user to re-wear the device. If the watch remains off for over three minutes,

it emits a louder alarm to attract public attention, providing details such as the user's appearance and clothing. Caregivers retain the ability to intervene and contact emergency services at any stage.

3.5 Comparison and Overview of Features

Existing solutions often sacrifice patient autonomy for safety. Our system balances both, integrating technology to enhance independence and security. Table 2 compares key features, highlighting our unique improvements.

4 LIMITATIONS

While the design reflects a thoughtful integration of expert insights and caregiver feedback, its practical effectiveness remains to be demonstrated in real-world settings. As this work is currently a design proposal, it has not yet been empirically validated through user testing with Alzheimer's patients or through comprehensive engagement with larger healthcare institutions and policymakers. Consequently, the system's usability, reliability, and broader ecological impact require further investigation in future research.

5 CONCLUSION AND FUTURE WORK

"Watch-Out" is a user-friendly safety system designed to address wandering in individuals with Alzheimer's, prioritizing both safety and dignity. Developed with insights from expert consultations and caregiver feedback, the system integrates an NFC-enabled smart door lock, a smartwatch, and real-time caregiver notifications. These features have demonstrated effectiveness in enhancing safety while preserving patient autonomy. Future efforts will focus on advancing "Watch-Out" from concept to widespread implementation, with plans to secure funding to ensure accessibility for a larger population. Collaborations with healthcare providers are being explored to refine its capabilities and support successful implementation.

"Watch-Out" seeks to redefine Alzheimer's care by blending safety with dignity, creating a future where individuals with Alzheimer's can navigate their world with confidence and caregivers can experience peace of mind knowing their loved ones are protected.

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